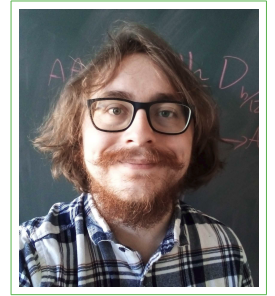


Owen Garnier

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Studies

- 2024 **PhD in mathematics**, LAMFA, Amiens.
Titre : Garside groupoids and complex braid groups.
Director : Ivan Marin
Other members of the Jury : Yves Guiraud, Jon McCammond, Adrien Brochier, Cédric Bonnafé, David Chataur, François Digne, Eddy Godelle.
- 2018–2021 **Masters degree in mathematics (Algebra, number theory and applications)**, Université de Picardie Jules Verne, Amiens.
With honors (Mention très bien).
Thesis: Théorie de Garside, applications aux groupes de tresses complexes. Cas de G_{31} . (supervisor: Ivan Marin).
- 2020 **Agrégation externe de mathématiques**, Rank: 10/387.
- 2018–2019 **Fourth year mathematics**, Université de Picardie Jules Verne, Amiens.
With honors (Mention très bien).
Thesis: Compléments sur l'homologie générale. Théorie des catégories (ex)triangulées. (supervisor: Yann Palu).
- 2015–2018 **Licence in mathematics**, Université de Picardie Jules Verne, Amiens.
With honors (Mention très bien).

Professional experiences

- 2026–2027 **Temporary research and teaching assistant (ATER)**, Université de Rennes, Rennes (France).
- 2025–2026 **Post-doctoral researcher**, University of Sevilla, Sevilla (Spain).
Supervised by María Cumplido Cabello.
- 2024–2025 **Temporary research and teaching assistant (ATER)**, Université de Picardie Jules Verne, Amiens (France).
- 2021–2024 **PhD student with teaching duties**, Université de Picardie Jules Verne, Amiens (France).
Supervised by Ivan Marin.

Distinctions and prizes

- 2025 **PhD thesis award of the Université de Picardie Jules Verne**, Award given to the two best PhD thesis in science, technology and health subjects defended in 2024

Research papers

- 2026 **Proof of Shvartsman's conjecture on braid groups of projective complex reflection groups.**
Proceedings of the American Mathematical Society 154, no. 10, p. 4077–4090
- 2026 **Springer categories for regular centralizers in well-generated complex braid groups.**
Advances in Mathematics 495, Paper No. 110971.
- 2024 **Uniqueness of roots up to conjugacy in circular and hosohedral-type Garside groups.**
Journal of Group Theory 27, 5 (2024), pp. 1091–1128.
- 2024 **Generalization of the Dehornoy-Lafont order complex to categories. Application to exceptional braid groups.**
Applied Categorical Structures 32, 1 (2024) Paper No. 1, 28 pp.
- 2023 **Regular theory in complex braid groups.**
Journal of Algebra 620 (2023), pp. 534–557.

Preprints

- 2025 **Generalized J -groups, J -braid groups and Seifert link groups, With I. Haladjian**
- 2025 **Conjugacy invariants and rigidity in Garside groups: a uniformity phenomenon, With M. Calvez, J. González-Meneses and B. Wiest**
Submitted to the *Journal of Algebra*
- 2025 **Normalisers of parabolic subgroups of Artin-Tits groups and Tits cone intersections, With E. Heng, A. Licata and O. Yacobi**
Submitted to *Geometry and Topology*
- 2024 **Parabolic subgroups of complex braid groups: the remaining case.**

Communications and Talks

Talks in conferences

- 2026 **Normalisers of parabolic subgroups in Coxeter groups and universal coverings,**
Combinatorics meet Coxeter Groups, Bielefeld (Germany).
- Seifert link groups and their torsion quotients,**
Workshop on geometric topology, Sevilla (Spain).
- 2025 **Sous-groupes paraboliques des trickle groups,**
Workshop on trickle groups and their applications, Tours (France).
- Système de réécritures et solutions au problème du mot,**
Workshop on trickle groups and their applications, Tours (France).
- Sous-groupes paraboliques des groupes de tresses complexes,**
Journées jeunes chercheurs et chercheuses en topologie, Amiens (France).
- Groupeïdes de Garside et groupes de tresses complexes,**
Colloque tournant du RT algèbre, Clermont-Ferrand (France).
- 2023 **Homology computations for complex braid groups with the Dehornoy-Lafont order complex,**
Conférence 2023 du GDR Théorie de l'Homotopie et Applications, Lille (France).
- Bessis braid category for the complex braid group B_{31} ,**
Winterbraids XII, Tours (France).
- 2022 **Regular theory in complex braid groups,**
Journées tresses 2022, Amiens (France).

Talks in seminars

- 2026 **Groupes de tresses complexes projectifs,**
Analytic geometry seminar, IRMAR (Rennes, France).
- Normalizers of parabolic subgroups in Artin and Coxeter groups,**
Algebra seminar, Dep. de Álgebra (Seville, Espagne).

- 2025 **Normalisateurs des sous-groupes paraboliques dans les groupes d'Artin et de Coxeter**,
Topology seminar, IMJ-PRG (Paris, France).
Normalizers of parabolic subgroups in Artin and Coxeter groups,
Group theory seminar, ICMAT (Madrid, Spain).
Projective complex braid groups,
Algebra and Topology seminar, ANU (Canberra, Australia).
Sous-groupes paraboliques, des tresses aux groupoïdes de Garside,
Geometry, Algebra, Dynamics and Topology seminar, IMB (Dijon, France).
Groupoïdes de Garside paraboliques et applications aux groupes de tresses complexes,
Algebra and Geometry seminar, LMNO (Caen, France).
- 2024 **Des partitions non-croisées à l'étude des sous-groupes paraboliques des groupes de tresses complexes**,
SPACE seminar, IDP (Tours, France).
Sous-groupes paraboliques et éléments réguliers des groupes de tresses complexes,
Darboux seminar, IMAG (Montpellier, France).
Homology of categories and the Dehornoy-Lafont order complex,
Seminar of the "Ouragan" team, IMJ-PRG (Paris, France).
- 2024 **Description de l'espace des orbites régulières d'un groupe de réflexions complexes**,
Topology seminar, LPP (Lille, France).
- 2023 **Étude Garside du groupe de tresses complexe $B(G_{31})$** ,
GATo Seminar, LAMFA (Amiens, France).
Garside study of the complex braid group $B(G_{31})$,
Groupes, representations and geometry seminar, IMJ-PRG (Paris, France).
Homology of a category and the Dehornoy-Lafont order complex,
Paris algebra seminar, IHP (Paris, France).
- 2022 **Regular theory in complex braid groups**,
Algebra seminar, Dep. de Álgebra (Seville, Espagne).
Théorie régulière dans les groupes de tresses complexes,
GATo seminar, LAMFA (Amiens, France).

Conferences

- 2027 **Winterbraids XVI**,
Rennes (France), from 30/03 to 02/04.
Forum on Artin & Coxeter groups: Examples & Theory in Sydney,
Sydney (Australia), from 01/02 to 26/02.
- 2026 **Buildings Workshop 2026**,
Bielefeld (Germany), from 05/10 to 08/10.
Combinatorics meet Coxeter Groups,
Bielefeld (Germany), from 30/09 to 02/10.
Workshop on geometric topology,
Seville (Spain), 19/02.
- 2025 **Workshop on trickle groups and their applications**,
Tours (France), from 25/06 to 27/06.
Artin groups, Braids and mapping class groups: Celebrating the work of Luis Paris,
Cáceres (Spain), from 16/06 to 20/06 .
Journées jeunes chercheurs et chercheuses en topologie,
Amiens (France), from 09/04 to 11/04.
Colloque tournant du RT algèbre,
Clermont-Ferrand (France), from 05/03 to 07/03.

Winterbraids XIV,

Bordeaux, from 03/02 to 06/02.

- 2023 **Conférence 2023 du GDR Théorie de l'Homotopie et Applications,**
Lille (France), from 24/10 to 27/10.

École d'été: À la découverte des correspondances de Langlands locales,
Amiens (France), du 28/08 au 01/09.

Workshop: Various Guises of Reflection Arrangements,
Edimbourg (UK), from 13/03 to 17/03.

Spring School on Real, Complex, and Symplectic Reflection Groups,
Bochum (Germany), from 06/03 to 10/03.

Winterbraids XII,

Tours, from 21/02 to 24/02.

- 2022 **Journées tresses 2022 : Groupes de tresses généralisés,**
Amiens (France), from 29/08 to 02/09.

- 2021 **Summer school: The dual approach to Coxeter and Artin groups; Garside theory and applications,**
Berlin (Germany), from 30/08 to 04/09.

Teaching

- 2026–2027
 - Mathematics for biology 1,2 (first year biology) (51h).
 - Analysis (University institute of technology) (20h).
 - Differential equations (second year mathematics) (30h).
 - Optimization (third year mathematics) (24h).
 - Preparation to the “agrégation externe” (M.Sc. mathematics) (40h).
- 2024–2025
 - Calculus (first year science) (30h).
 - Set theory (second year mathematics) (20h).
 - Groups, Rings, Fields (third year mathematics) (36h).
 - Linear algebra 1 (first year science) (32h).
 - parametrized curves (first year mathematics) (36h).
 - Complex analysis (third year mathematics) (20h).
- 2023–2024
 - Advanced linear algebra (third year mathematics) (20h).
 - Complex analysis (third year mathematics) (20h).
 - Preparation to the “agrégation externe” (M.Sc. mathematics) (24h).
- 2022–2023
 - Advanced linear algebra (third year mathematics) (20h).
 - Algebraic algorithmics (third year mathematics) (20h).
 - Preparation to the “agrégation externe” (M.Sc. mathematics) (24h).
- 2021–2022
 - Calculus (first year science) (24h).
 - Advanced linear algebra (third year mathematics) (20h).
 - Complex numbers and geometry (third year mathematics) (20h).

Other activities

- 2025 **Co-organizer of the workshop on trickle groups and their applications,** Tours.
With Igor Haladjian.
- 2023–2024 **Co-organizer of the PhD student seminar of the LAMFA,** Amiens.
With Nicolas Beuvin et Michaël Schoonheere.
- 2022–2023 **Co-organizer of the PhD student seminar of the LAMFA,** Amiens.
With Felipe Arbulú et Ismaïl Razack.
- 2022 **Research stay in Sevilla,** *Universidad de Sevilla*, Sevilla (Spain).
Work with Juan Gonzalez-Meneses (3 weeks).
- 2022 **Co-organizer of the braid meeting 2022,** Amiens.
With Ivan Marin and Karine Sorlin.